OPERATION RESEACH CHAPTER 4 MCQs

Section 4.1: The Essence of the Simplex Method

1. Who developed the simplex method and in what year?

   A. George Dantzig in 1947
   B. Narendra Karmarkar in 1984
   C. John von Neumann in 1945
   D. Leonid Kantorovich in 1939

2. Explain the geometric relationship between constraint boundaries and corner-point solutions.

   A. Corner-point solutions lie at the intersection of constraint boundaries.
   B. Corner-point solutions are strictly interior to the constraint boundaries.
   C. Constraint boundaries are always parallel to corner-point solutions.
   D. Corner-point solutions represent the slopes of constraint boundaries.

3. What distinguishes a corner-point feasible (CPF) solution from a corner-point infeasible solution?

   A. CPF satisfies all constraints; infeasible violates at least one.
   B. CPF violates nonnegativity; infeasible violates functional constraints.
   C. CPF is inside the region; infeasible is on the boundary.
   D. CPF has slack variables equal to 0; infeasible has artificial variables equal to 0.

4. Define what it means for two CPF solutions to be adjacent in a model with n decision variables.

   A. They share $n - 1$ constraint boundaries.
   B. They share exactly 1 constraint boundary.
   C. They share n constraint boundaries.
   D. They share 0 constraint boundaries.

5. How does the optimality test for the simplex method utilize adjacent CPF solutions?

   A. If no adjacent CPF solution is better, the current one is optimal.
   B. If all adjacent CPF solutions are worse, the origin is optimal.

C. It checks all CPF solutions simultaneously against the origin.

D. It averages the Z values of all adjacent CPF solutions.

6. Describe the initialization step of the simplex method from a geometric viewpoint.

A. Choose the origin as the initial CPF solution.

B. Choose the centroid of the feasible region.

C. Choose any corner-point infeasible solution.

D. Move along the x_1 axis until a boundary is hit.

7. Why is the origin commonly chosen as the initial CPF solution when all nonnegativity constraints apply?

A. No calculations are required to identify this CPF solution.

B. It is always the optimal solution.

C. All slack variables are zero at the origin.

D. It automatically satisfies all equality constraints.

8. If an edge of the feasible region provides a positive rate of improvement in Z, what does this imply about the adjacent CPF solution?

A. The adjacent CPF solution is better than the current one.

B. The adjacent CPF solution is worse than the current one.

C. The current CPF solution is already optimal.

D. The problem has an unbounded feasible region.

9. According to the key solution concepts, why does the simplex method only examine CPF solutions instead of interior points?

A. Finding one optimal solution requires only finding a best CPF solution.

B. Interior points are computationally unstable.

C. The feasible region only contains CPF solutions.

D. Fractional solutions are not allowed in linear programming.

10. How does the simplex method determine when an optimal solution has been reached geometrically?

A. No edges emanating from the current CPF give a positive rate of improvement in Z.

B. The basic variables are all strictly equal to zero.

C. The origin is reached after circling the feasible region.

D. The feasible region is proven to be unbounded.

Section 4.2: Setting up the Simplex Method

11. What is the purpose of introducing slack variables into a linear programming model?

    A. To convert inequality constraints to equality constraints.
    B. To eliminate artificial variables from the model.
    C. To artificially increase the objective function value.
    D. To make corner-point infeasible solutions feasible.

12. Convert the inequality $3x_1 + 2x_2 \leq 18$ into an equality constraint using a slack variable.

    A. $3x_1 + 2x_2 + x_3 = 18$
    B. $3x_1 + 2x_2 - x_3 = 18$
    C. $3x_1 + 2x_2 + x_3 \leq 18$
    D. $3x_1 + 2x_2 = 18 + x_3$

13. What is the difference between the original form and the augmented form of a linear programming model?

    A. The augmented form includes supplementary variables like slack variables.
    B. The original form has equality constraints only.
    C. The augmented form has only nonnegativity constraints.
    D. The original form includes surplus variables.

14. Define an augmented solution in the context of the simplex method.

    A. A solution for original variables augmented by corresponding slack variables.
    B. A solution with only basic variables.
    C. A solution ignoring nonnegativity constraints.
    D. A geometric representation of constraints in n-dimensional space.

15. What is a basic solution, and how is it derived from an augmented corner-point solution?

    A. It is the algebraic equivalent of an augmented corner-point solution.
    B. It is any point in the interior of the feasible region.
    C. It is the origin before augmenting variables.
    D. It is a solution containing negative basic variables.

16. What distinguishes a basic feasible (BF) solution from a basic infeasible solution?

    A. A BF solution satisfies all nonnegativity constraints.
    B. A BF solution has no slack variables.
    C. An infeasible solution has $Z = 0$.
    D. A BF solution has unconstrained variables.

17. If a linear programming problem has 5 variables and 3 functional constraints, how many degrees of freedom are there when solving the system of equations?

A. 2
B. 3
C. 5
D. 8

18. In a basic solution, what value is arbitrarily assigned to the nonbasic variables?

A. They are set equal to zero.
B. They are set equal to the right-hand side.
C. They are set equal to their objective coefficients.
D. They are set equal to M.

19. State the property that determines if two basic feasible solutions are adjacent algebraically.

A. All but one of their nonbasic variables are the same.
B. They share all nonbasic variables.
C. They have identical basic variables with different values.
D. Both have an objective value of $Z = 0$.

20. How does switching one variable from nonbasic to basic (and vice versa) relate to moving between adjacent BF solutions?

A. It algebraically represents moving between adjacent BF solutions.
B. It converts a basic solution into an artificial problem.
C. It changes the coefficients of the objective function.
D. It makes the problem unbounded.

Sections 4.3 & 4.4: The Algebra and Tabular Form of the Simplex Method

21. Why is the initial BF solution easily obtained when all functional constraints are $\leq$ with nonnegative right-hand sides?

A. Slack variables can serve as the initial basic variables.
B. Artificial variables are zero by default.
C. The origin is always the final optimal solution.
D. There are no nonbasic variables in this form.

22. Describe the algebraic procedure used to identify the entering basic variable in Step 1 of an iteration.

A. Select the nonbasic variable with the 'most negative' coefficient in Row 0.
B. Select the basic variable with the largest positive coefficient.
C. Select the variable with the smallest ratio in the minimum ratio test.
D. Select the variable with a zero coefficient in Row 0.

23. What is the purpose of the minimum ratio test in Step 2 of a simplex iteration?

A. To determine the leaving basic variable.
B. To determine the entering basic variable.
C. To calculate the maximum optimal Z.
D. To identify if the feasible region is bounded.

24. If a basic variable has a zero or negative coefficient in the pivot column, why is it excluded from the minimum ratio test?

A. It would not decrease as the entering variable increases, thus won't hit zero first.
B. It leads to division by zero only.
C. It indicates the presence of multiple optimal solutions.
D. It means the constraint is already strictly binding.

25. Explain the operations used in Gaussian elimination during step 3.

A. Multiply/divide an equation by a constant, and add/subtract a multiple of one equation to another.
B. Eliminate the leaving basic variable from the objective function only.
C. Add a penalty M to all constraint coefficients.
D. Set all nonbasic variables to 1 and solve.

26. What does "proper form from Gaussian elimination" mean in the context of a simplex tableau?

A. Each basic variable has a +1 coefficient in its equation and 0 in all others.
B. All coefficients in the tableau are strictly positive.
C. Only nonbasic variables have +1 coefficients.
D. Row 0 contains no variables and only constants.

27. In tabular form, what information is recorded in "Row 0" of the simplex tableau?

A. The coefficients of the objective function equation.
B. The right-hand side limits for the functional constraints.
C. The artificial penalties applied to slack variables.
D. The current values of the basic variables.

28. How is the leaving basic variable identified visually in the simplex tableau?

A. By finding the pivot row derived from the minimum ratio test.
B. By locating the pivot column from Row 0.
C. By finding the most negative value in Row 0.
D. By locating a zero in the rightmost column.

29. What happens to the objective function value Z when the entering basic variable is increased?

A. Z increases at a rate given by the negative of its coefficient in Row 0.
B. Z decreases proportionally to the leaving variable.
C. Z remains constant until the final step is reached.

D. Z is instantly forced to zero.

30. Convert the objective function $Z = 3x_1 + 5x_2$ into the equality form required for Row 0 of the initial simplex tableau.

A. $Z - 3x_1 - 5x_2 = 0$

B. $Z + 3x_1 + 5x_2 = 0$

C. $-Z - 3x_1 - 5x_2 = 0$

D. $Z = 0$

Section 4.5: Tie Breaking in the Simplex Method

31. How does the simplex method resolve a tie for the entering basic variable?

A. The selection between the tied variables may be made arbitrarily.

B. It always chooses the variable with the smallest index.

C. It stops, as the problem has multiple optimal solutions.

D. It chooses the variable with the largest minimum ratio.

32. What is degeneracy in linear programming?

A. Basic variables having a value of exactly zero.

B. Nonbasic variables having a value of zero.

C. All coefficients in Row 0 being positive.

D. The feasible region being unbounded.

33. How can a tie for the leaving basic variable lead to a degenerate basic variable?

A. The tied basic variables reach zero simultaneously, making the one not chosen zero in the next solution.

B. The entering variable becomes negatively valued.

C. The objective function drastically decreases.

D. Slack variables are forced to become artificial variables.

34. Explain the theoretical risk of a perpetual loop (cycling) in the simplex method due to degeneracy.

A. Z remains unchanged at each iteration, potentially repeating the same sequence of solutions indefinitely.

B. The entering variable increases indefinitely without bound.

C. The origin is continuously revisited after every 3 iterations.

D. Artificial variables can never leave the basis once degenerate.

35. What is the significance of having no leaving basic variable (no positive coefficients in the pivot column) in step 2 of an iteration?

A. It indicates that the objective function Z is unbounded.

B. It means the problem has no feasible solutions.

C. It indicates that the current solution is optimal.

D. It proves the problem is highly degenerate.

36. If the simplex method indicates that Z is unbounded, what does this usually imply about the real-world problem formulation?

A. A mistake was made, such as omitting relevant constraints.

B. The company is guaranteed to make infinite profit.

C. Nonnegativity constraints are completely unnecessary.

D. The optimal solution is exactly zero.

37. How can you detect the existence of multiple optimal solutions from the final simplex tableau?

A. At least one nonbasic variable has a zero coefficient in Row 0.

B. All Row 0 coefficients are strictly positive.

C. A basic variable has a negative value in the right column.

D. $Z = 0$ in the final iteration.

38. If a problem has multiple optimal solutions, how are the other optimal BF solutions found using the tableau?

A. Perform extra iterations by choosing a nonbasic variable with a zero Row 0 coefficient as the entering variable.

B. Re-solve the problem from scratch using the Big M method.

C. Change the objective function to minimize instead of maximize.

D. Eliminate the slack variables and pivot the artificial variables.

39. Define a convex combination of solutions.

A. A weighted average of solutions where the weights are nonnegative and sum to 1.

B. The absolute sum of all CPF solutions.

C. The exact geometric center of the feasible region.

D. The cross-product of basic and nonbasic variables.

40. If (2, 6) and (4, 3) are optimal CPF solutions, express the set of all optimal solutions algebraically.

A. $w_1(2, 6) + w_2(4, 3)$ where $w_1, w_2 \geq 0$ and $w_1 + w_2 = 1$

B. $(2 + 4, 6 + 3)/2$

C. $(2, 6) + (4, 3)$

D. $w_1(2, 6) - w_2(4, 3)$

Section 4.6: Adapting to Other Model Forms

41. What is the standard approach used to find an initial BF solution when the model has equality constraints?

    A. The artificial-variable technique.
    B. Multiplying the constraint by -1.
    C. Removing the constraint entirely.
    D. Using only surplus variables.

42. Explain the purpose of introducing an artificial variable into an equality constraint.

    A. To serve as a convenient initial basic variable for that equation.
    B. To artificially increase the objective function value.
    C. To represent unused resources in the constraint.
    D. To convert $\leq$ constraints to $\geq$ constraints.

43. In the Big M method, why is an overwhelming penalty assigned to the artificial variable in the objective function?

    A. To force the artificial variable to become zero in the optimal solution.
    B. To maximize the artificial variable's value.
    C. To prevent the slack variable from entering the basis.
    D. To balance the units of the equations.

44. What happens if an artificial variable remains strictly greater than zero in the final optimal solution?

    A. The real problem has no feasible solutions.
    B. The real problem has multiple optimal solutions.
    C. The penalty M used was too small.
    D. The objective function is unbounded.

45. How does the artificial-variable technique handle functional constraints in $\geq$ form?

    A. Introduce a surplus variable (subtracted) and an artificial variable (added).
    B. Add a single slack variable.
    C. Multiply the entire constraint by M.
    D. Change the constraint to $\leq$ by replacing the variables.

46. What is a surplus variable and how does it differ from a slack variable?

    A. Surplus subtracts excess over RHS to form equality; Slack adds to reach RHS.
    B. Surplus is always negative; Slack is always positive.
    C. Surplus is used for $\leq$ constraints; Slack is used for $\geq$ constraints.
    D. Surplus adds an artificial penalty; Slack does not.

47. Convert the constraint $0.6x_1 + 0.4x_2 \geq 6$ into an equality using surplus and artificial variables.

A. $0.6x_1 + 0.4x_2 - x_5 + \bar{x}_6 = 6$

B. $0.6x_1 + 0.4x_2 + x_5 = 6$

C. $0.6x_1 + 0.4x_2 - \bar{x}_6 = 6$

D. $0.6x_1 + 0.4x_2 + x_5 - \bar{x}_6 = 6$

48. How is a minimization problem converted into an equivalent maximization problem for the simplex method?

A. Maximize $-Z = -\sum c_j x_j$

B. Minimize $-Z = \sum c_j x_j$

C. Maximize $Z = \sum -c_j x_j + M$

D. Take the mathematical reciprocal of Z.

49. Why must basic variables be algebraically eliminated from Row 0 before starting the simplex method with artificial variables?

A. So the negative of the coefficient gives the correct rate of increase for Z.

B. To make Z equal to zero initially.

C. Because artificial variables cannot exist in Row 0.

D. To satisfy the nonnegativity constraints automatically.

50. Describe the overall two phases of the Two-Phase Method.

A. Phase 1: Minimize artificial variables to find a feasible solution; Phase 2: Solve the real problem.

B. Phase 1: Maximize Z; Phase 2: Minimize Z.

C. Phase 1: Use the Big M method; Phase 2: Use the small M method.

D. Phase 1: Solve graphically; Phase 2: Solve algebraically.

51. In Phase 1 of the Two-Phase Method, what is the objective function being minimized?

A. Minimize $Z = \Sigma$ artificial variables

B. Maximize $Z = \Sigma$ original variables

C. Minimize $Z = \Sigma$ slack variables

D. Maximize $Z = M$

52. If Phase 1 concludes with a minimum $Z = 0$, what does this indicate for Phase 2?

A. A basic feasible solution for the real problem has been found.

B. The real problem has no feasible solutions.

C. The problem is strictly unbounded.

D. The origin is the optimal solution for Phase 2.

53. Why is the Two-Phase method often preferred over the Big M method in computer codes?

A. It avoids numerical instability caused by assigning a huge numerical value to M.

B. It always requires exactly half the iterations.

C. It does not require the use of artificial variables.

D. The Big M method cannot handle $\geq$ constraints.

54. How is a decision variable with a lower bound constraint $x_j \geq -10$ converted to a non-negative variable?

A. Substitute $x_j' = x_j + 10$ where $x_j' \geq 0$

B. Substitute $x_j' = x_j - 10$ where $x_j' \geq 0$

C. Substitute $x_j = x_j^+ - x_j^-$

D. Multiply the constraint by -1.

55. If a variable x_j has no lower bound, how is it substituted in the model to satisfy the non-negativity restriction?

A. $x_j = x_j^+ - x_j^-$ where $x_j^+, x_j^- \geq 0$

B. $x_j = |x_j|$

C. $x_j = x_j' + M$

D. The variable is dropped from the model completely.

56. If x_j is replaced by $x_j^+ - x_j^-$, why must at least one of these new variables be zero in any BF solution?

A. Only endpoints (where one is zero) are corner-point solutions on the substitution line.

B. They mathematically cancel each other out.

C. The simplex method explicitly forces both to zero.

D. Artificial variables replace them during the iterations.

57. What modification can be made to minimize the number of new variables when multiple variables lack lower bounds?

A. Replace each x_j by $x_j' - x''$ where x'' is a single variable shared among all.

B. Only use x_j^+ for all variables.

C. Assume all lower bounds are zero.

D. Use Phase 1 to independently find the bounds.

58. Using the Big M method for a minimization problem, what sign should the penalty term M have in the objective function?

A. $+M$

B. $-M$

C. 0

D. M^2

59. Graphically, how does the feasible region of the artificial problem compare to the real problem?

A. The artificial region is larger and encompasses the real region.

B. The artificial region is strictly smaller than the real region.

C. They are completely identical regions.

D. The artificial region is un-graphable.

60. What is the result of applying the simplex method if a problem has no feasible solutions?

A. The final solution has at least one artificial variable > 0.

B. The minimum ratio test fails completely.

C. Z approaches infinity without bound.

D. All nonbasic variables become negative.

Sections 4.7, 4.8, & 4.9: Postoptimality, Implementation, & Interior-Point Approach

61. What is the purpose of reoptimization in postoptimality analysis?

A. To deduce how model changes affect the final tableau to quickly find a new optimal solution.

B. To solve the problem from scratch to verify the results.

C. To graphically visualize the new constraints.

D. To systematically introduce new artificial variables.

62. Why is reoptimization much more efficient than solving a modified model from scratch?

A. The new optimal solution is usually close to the prior optimal, requiring few iterations.

B. It skips the minimum ratio test entirely.

C. It utilizes interior-point methods instead.

D. It eliminates all nonbasic variables immediately.

63. Define the shadow price of a resource in linear programming.

A. The marginal value/rate at which Z increases by slightly increasing resource b_i.

B. The cost of producing one unit of product.

C. The penalty M applied to artificial variables.

D. The numerical value of the slack variable.

64. How are shadow prices identified from the final simplex tableau?

A. The coefficient of the ith slack variable in Row 0.

B. The right-hand side of the ith constraint.

C. The coefficient of the original variables in Row 0.

D. The pivot number of the final iteration.

65. If a constraint $x_1 \leq 4$ is not binding at the optimal solution, what is the shadow price of this resource?

A. Zero

B. Positive

C. Negative

D. *M*

66. Explain the economic interpretation of a positive shadow price for a resource.

A. The resource is a scarce good; increasing it increases profit.

B. The resource is a free good; increasing it has no effect.

C. The resource is currently overpriced.

D. The constraints are improperly formulated.

67. What does it mean for a parameter to be “sensitive” in the context of sensitivity analysis?

A. It is a parameter whose value cannot be changed without changing the optimal solution.

B. It is a parameter that must be negative.

C. It is a parameter associated only with artificial variables.

D. It is a parameter with a shadow price of zero.

68. Define the “allowable range to stay optimal” for an objective function coefficient.

A. The range of values over which the current optimal solution remains optimal.

B. The range over which the constraint remains binding.

C. The range over which the feasible region is bounded.

D. The range where artificial variables are strictly positive.

69. What is the “allowable range to stay feasible” for a right-hand side value?

A. The range over which the current optimal BF solution (adjusted) remains feasible.

B. The range where the objective function value does not change.

C. The range where no slack variables are needed.

D. The range where shadow prices are strictly zero.

70. How does parametric linear programming differ from standard sensitivity analysis?

A. Parametric studies simultaneous changes over a range; Sensitivity changes one parameter at a time.

B. Parametric only applies to interior-point methods.

C. Sensitivity is graphical; Parametric is strictly algebraic.

D. They are completely identical concepts.

71. What is the “revised simplex method” and why is it used in computer implementations?

A. A matrix form that computes and stores only the numbers needed for the current iteration.

B. A method that uses interior points instead of corner points.

C. A method that solves maximization problems without slack variables.

D. A graphical technique designed for exactly 3 variables.

72. For large problems, the computation time of the simplex method tends to be proportional to the cube of what parameter?

    A. The number of ordinary functional constraints.
    B. The number of decision variables.
    C. The number of artificial variables.
    D. The value of M used in the Big M method.

73. What is the typical rule of thumb for the number of iterations required by the simplex method?

    A. Roughly twice the number of functional constraints.
    B. Equal to the number of variables squared.
    C. Half the number of slack variables.
    D. 2^n where n is number of variables.

74. Who discovered the interior-point approach to solving linear programming problems in 1984?

    A. Narendra Karmarkar
    B. George Dantzig
    C. John von Neumann
    D. Leonid Khachiyan

75. How do the trial solutions of an interior-point algorithm differ from those of the simplex method?

    A. They are points strictly inside the boundary, not on corner points.
    B. They are strictly outside the feasible region.
    C. They only move along the outer edges.
    D. They do not satisfy nonnegativity constraints.

76. Compare the worst-case computational complexity of the simplex method and Karmarkar's algorithm.

    A. Simplex is exponential time; Karmarkar is polynomial time.
    B. Simplex is polynomial time; Karmarkar is exponential time.
    C. Both are exponential time algorithms.
    D. Both are polynomial time algorithms.

77. For huge linear programming problems, why might an interior-point algorithm require significantly fewer iterations than the simplex method?

    A. It shoots through the interior, bypassing the astronomical number of CPF solutions on the boundary.
    B. It calculates all constraints simultaneously in parallel.
    C. It simply ignores all $\geq$ constraints.
    D. It sets all nonbasic variables to zero instantly.

78. What is a key disadvantage of the interior-point approach regarding postoptimality analysis?

    A. It has limited capability for performing postoptimality analysis.
    B. It cannot handle minimization problems at all.
    C. It is much slower on huge problems.
    D. It cannot be implemented on standard computer software.

79. How can the simplex method be combined with the interior-point approach to overcome analytical drawbacks?

    A. Use interior-point to get near optimal, then switch to simplex to find the exact CPF and do postoptimality.
    B. Run both simultaneously and average the final answers.
    C. Use simplex for Phase 1 and interior-point for Phase 2.
    D. Multiply the final shadow prices from both methods.

80. Why is sparsity in the table of constraint coefficients beneficial for computer implementations of the simplex method?

    A. A high proportion of zero coefficients greatly accelerates computation per iteration.
    B. It prevents degeneracy from occurring.
    C. It eliminates the need for artificial variables.
    D. It guarantees a single unique optimal solution.

Part II: Numerical Questions & Tableaux

81. Consider the initial simplex tableau below for a maximization problem. Identify the entering basic variable, the leaving basic variable, and the pivot number.

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	Right Side
Z	(0)	1	-3	-5	0	0	0	0
x_3	(1)	0	1	0	1	0	0	4
x_4	(2)	0	0	2	0	1	0	12
x_5	(3)	0	3	2	0	0	1	18

82. After one iteration of the simplex method for the problem in Question 81, the following tableau is obtained. Is this solution optimal? If not, identify the next entering and leaving basic variables.

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	Right Side
Z	(0)	1	-3	0	0	$\frac{5}{2}$	0	30
x_3	(1)	0	1	0	1	0	0	4
x_2	(2)	0	0	1	0	$\frac{1}{2}$	0	6
x_5	(3)	0	3	0	0	-1	1	6

83. Consider the final optimal simplex tableau below. What are the optimal values of the decision variables x_1 and x_2, the slack variables x_3, x_4, x_5, and the objective function Z?

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	Right Side
Z	(0)	1	0	0	0	$\frac{3}{2}$	1	36
x_3	(1)	0	0	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2
x_2	(2)	0	0	1	0	$\frac{1}{2}$	0	6
x_1	(3)	0	1	0	0	$-\frac{1}{3}$	$\frac{1}{3}$	2

84. Based on the optimal tableau in Question 83, what are the shadow prices for resources 1, 2, and 3 (corresponding to slack variables x_3, x_4, x_5)?

85. In the tableau below, demonstrate why the problem has an unbounded objective function according to the minimum ratio test.

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	Right Side
Z	(0)	1	-3	-5	0	0
x_3	(1)	0	1	0	1	4

86. The following final tableau represents a problem with multiple optimal solutions. Which nonbasic variable can enter the basis without changing the value of Z?

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	Right Side
Z	(0)	1	0	0	0	1	0	18
x_1	(1)	0	1	0	$-\frac{1}{3}$	1	$\frac{2}{3}$	4
x_3	(2)	0	0	0	1	$\frac{1}{3}$	$-\frac{1}{3}$	2
x_2	(3)	0	0	1	$\frac{1}{2}$	0	0	6

87. Write the initial augmented equations (including slack variables) for the following problem: Maximize $Z = 2x_1 + 3x_2$ subject to $x_1 + x_2 \leq 4$ and $x_1, x_2 \geq 0$.

Answers

1.A 2.A 3.A 4.A 5.A 6.A 7.A 8.A 9.A 10.A 11.A 12.A 13.A 14.A 15.A 16.A 17.A 18.A

19.A 20.A 21.A 22.A 23.A 24.A 25.A 26.A 27.A 28.A 29.A 30.A 31.A 32.A 33.A 34.A

35.A 36.A 37.A 38.A 39.A 40.A 41.A 42.A 43.A 44.A 45.A 46.A 47.A 48.A 49.A 50.A

51.A 52.A 53.A 54.A 55.A 56.A 57.A 58.A 59.A 60.A 61.A 62.A 63.A 64.A 65.A 66.A

67.A 68.A 69.A 70.A 71.A 72.A 73.A 74.A 75.A 76.A 77.A 78.A 79.A 80.A